

DATA ANALYTICS · RESEARCH WING · POLICE DEPARTMENT

Crime Analysis

An interactive Tableau dashboard translating raw incident records into insight for crime prevention and resourcing.

2,47,797

INCIDENTS ANALYZED

2017-2022

REPORTING PERIOD

33

CRIME CATEGORIES

4

LINKED DASHBOARDS

Why Crime Analysis Matters

Crime analysis underpins law-and-order maintenance. By surfacing patterns in past criminal activity, it gives police departments and city agencies the evidence base they need to plan prevention programs, allocate patrols, and respond faster to emerging hotspots.

As a data analyst in the department's research wing, this project turns six years of incident-level records into a Tableau dashboard built for a public-facing communication website — prioritizing clear, honest data storytelling.

CRIME DATA ANALYSIS SUPPORTS

1

Criminal Investigation

Surfacing patterns that point investigators toward likely times, places, and methods.

2

Apprehension & Prosecution

Arrest-outcome data that helps evaluate case-closure performance.

3

Prevention Strategy

Hotspot and time-block trends that guide where prevention programs are targeted.

4

Resource Planning

Patrol and staffing decisions grounded in when and where incidents cluster.

Four Dashboards, One Story

The task: build an interactive Tableau dashboard/story for the department's communication website — covering where and what crime occurs, when it occurs, how it is trending, and how outcomes compare — all filterable by incident type and location.

01

Overall Crime Statistics

Citywide counts and types, a live incident feed, and a geo-map of recent activity.

02

Time Period Analysis

Incident distribution by weekday and hour, and share of reports by time block.

03

Trend Analysis

Year-over-year crime-rate change, compared across the same dates and times.

04

Comparative Analysis

Arrest vs. no-arrest outcomes and the share of incidents that are domestic.

Every dashboard shares interactive filters for Primary Type and Location Description, so users can drill into a specific crime or neighborhood.

THE DATA

Dataset at a Glance

2,47,797

TOTAL INCIDENT RECORDS

2017–2022

YEARS COVERED

33

PRIMARY CRIME TYPES

20%

ENDED IN AN ARREST

KEY FIELDS USED IN THE ANALYSIS

● Date & time

when each incident occurred

● Location Description

setting — street, residence, airport, etc.

● Beat, District, Ward, Community Area

police geography

● Primary Type / Description

the UCR-linked crime category

● Arrest / Domestic

outcome and domestic-incident flags

● Latitude / Longitude

point-level mapping

Overall Crime Statistics

2,47,797

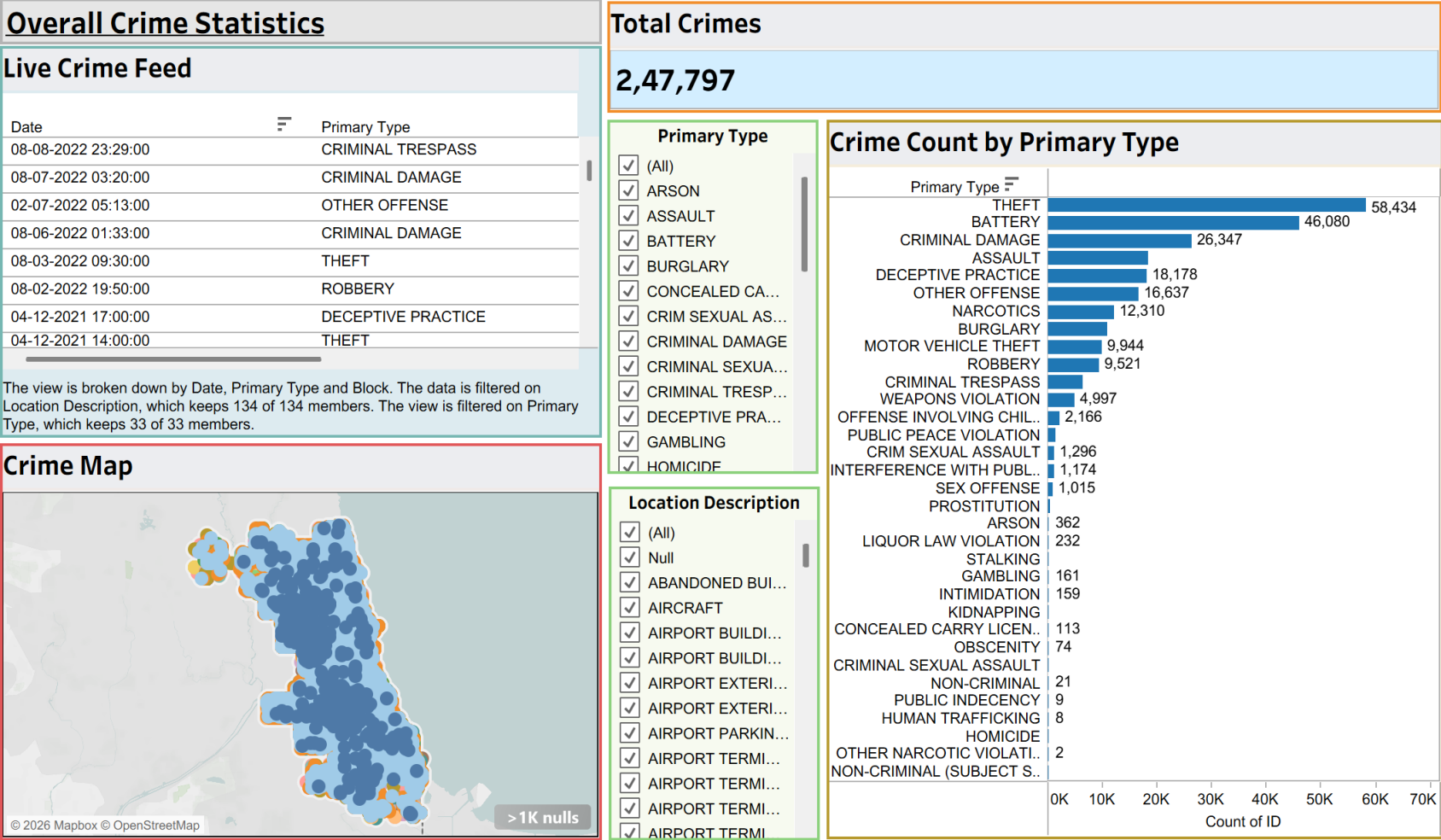
TOTAL CRIMES REPORTED

Theft

MOST COMMON INCIDENT — 58,434 CASES

WHAT IT SHOWS

- A live crime feed lists the most recent incidents with date, type, and block-level location.
- Theft and Battery alone account for roughly 42% of all reported crime.
- The geo-map plots incidents across the city, concentrated in the dense central and near-north areas.
- Primary Type and Location Description filters let staff isolate any category or neighborhood.



Time Period Analysis

12PM & 6-7PM

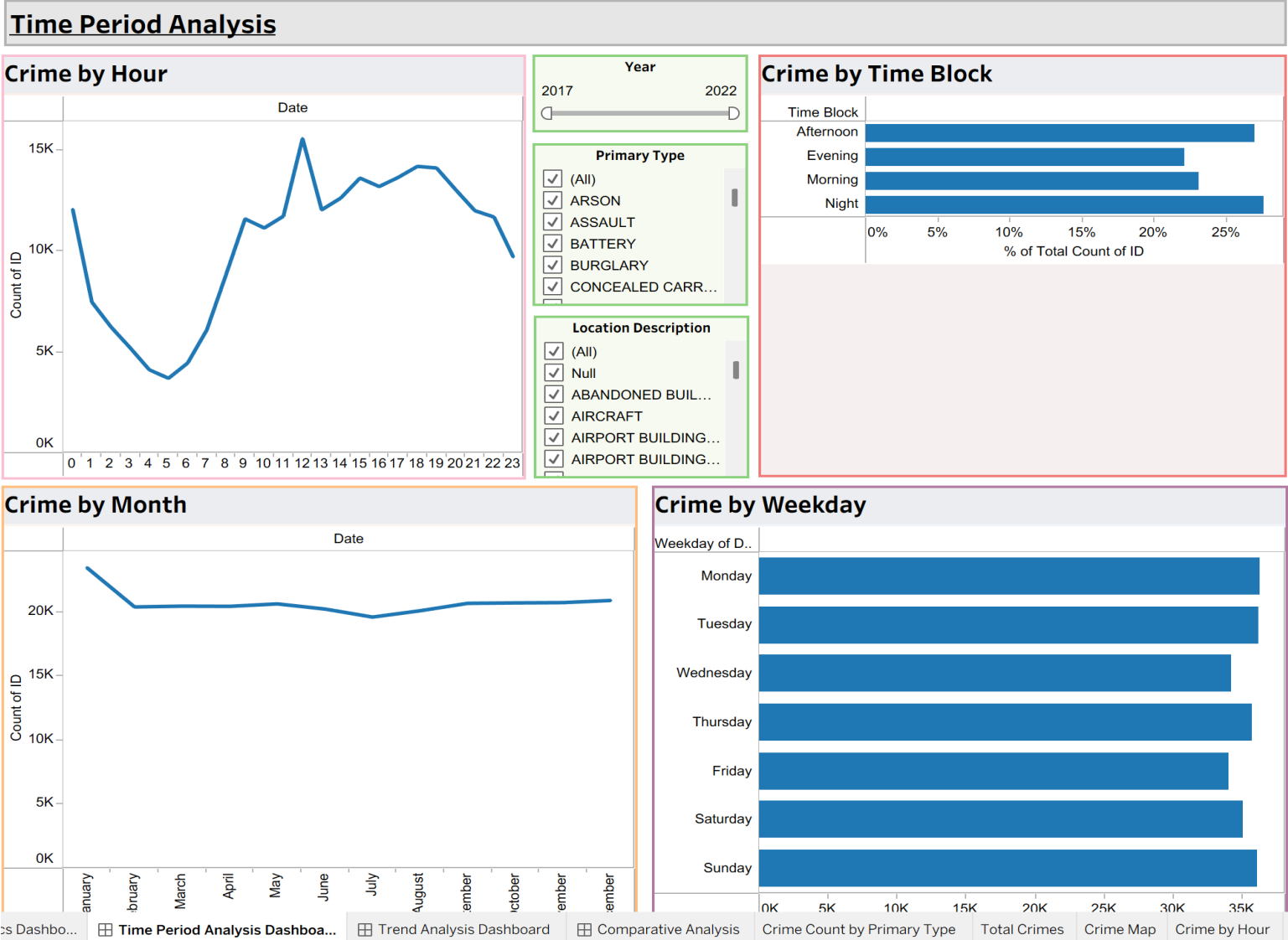
SHARPEST REPORTING PEAKS

27%

OF INCIDENTS OCCUR AT NIGHT

WHAT IT SHOWS

- Crime by Hour dips to its lowest near 5 AM (~3.7K) and climbs sharply from 8 AM, spiking at midday.
- Afternoon (27%) and Night (27%) together account for over half of all reported incidents.
- Weekday volumes are fairly even, with Wednesday and Friday slightly lower than the rest.
- Monthly volume is highest in January and dips modestly through the summer months.



Trend Analysis

2017-2018

PEAK YEARS — ~102K/YR

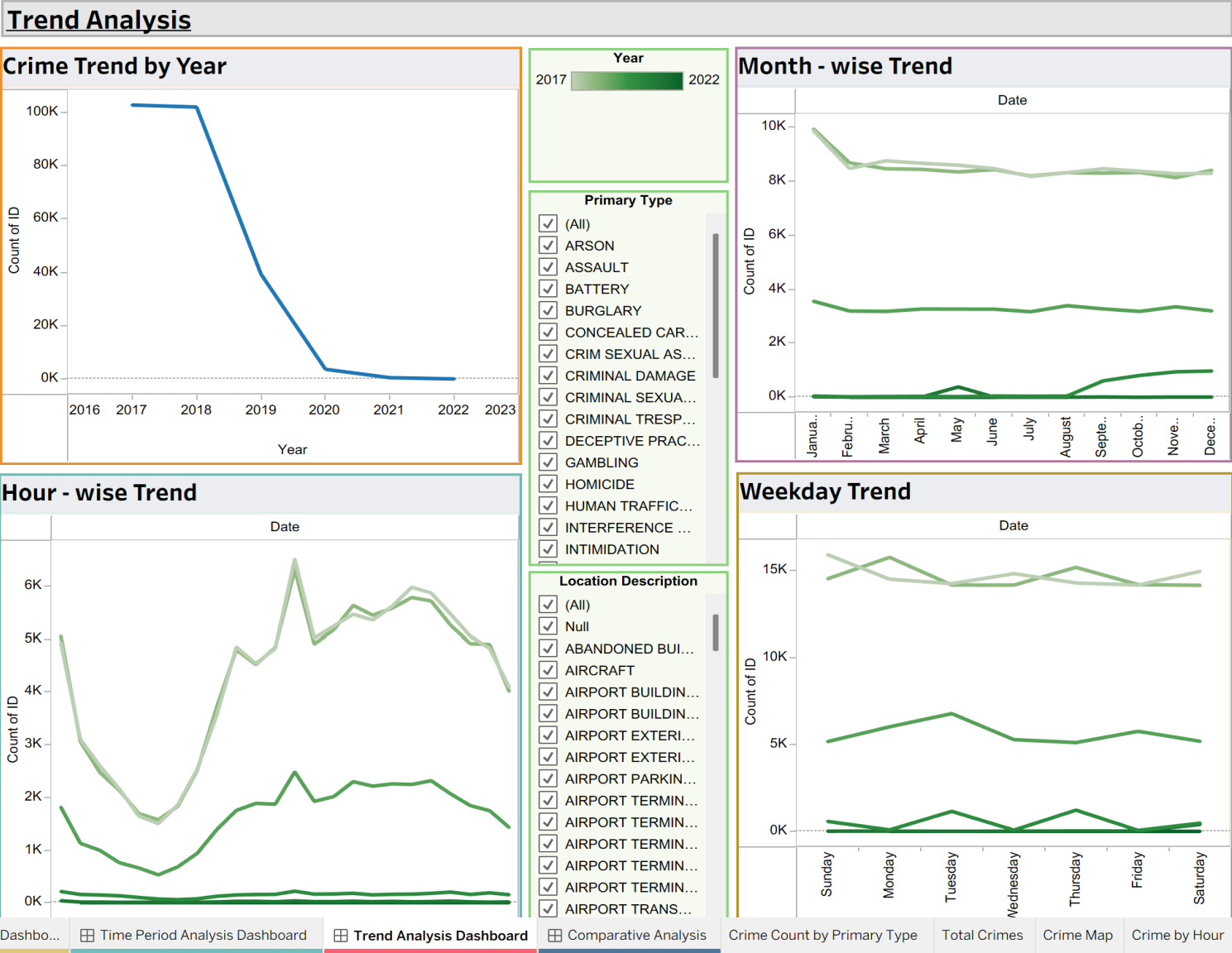
2019+

SHARP REPORTED DECLINE

WHAT IT SHOWS

- Annual crime count holds near 102K through 2017–2018, then falls sharply from 2019 onward.
- Month-, hour-, and weekday-wise trend lines let users compare shape of reporting across years.
- Hour-of-day and weekday shape stay broadly consistent year to year even as overall volume shifts.

Note: the drop after 2019 tracks with reduced record coverage in the source extract for later years, and should be read as a data-completeness effect rather than confirmed real-world decline.



Comparative Analysis

80% / 20%

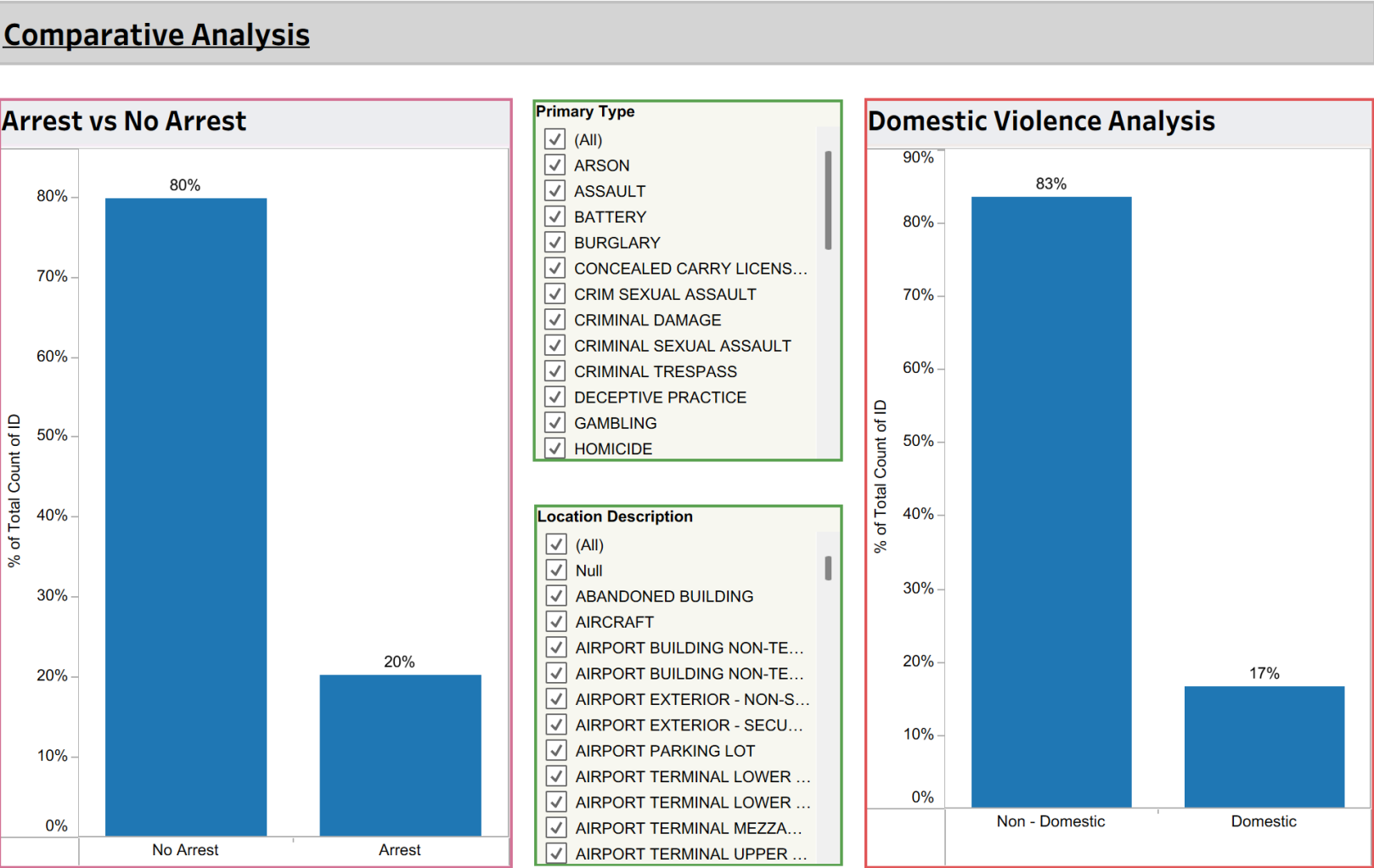
NO-ARREST VS. ARREST SPLIT

17%

OF INCIDENTS ARE DOMESTIC

WHAT IT SHOWS

- Four in five reported incidents in this dataset closed without an arrest.
- 83% of incidents are non-domestic; domestic incidents make up 17% of the total.
- Both views share the Primary Type and Location Description filters for category-level drill-down.



What the Dashboard Tells the Department

1

Theft & Battery dominate

Together they make up ~42% of all reports — the highest-leverage categories for prevention spend.

2

Afternoons and nights peak

Reporting concentrates in the afternoon and night blocks — the strongest case for shift-based patrol weighting.

3

Low arrest resolution

Only 20% of incidents end in arrest, a starting point for reviewing investigative capacity.

4

Domestic cases are a minority

17% of incidents are domestic, but likely warrant a dedicated response protocol given their nature.

5

Reporting is fairly steady weekly

Day-of-week volume varies little, so staffing doesn't need heavy weekday-to-weekday swings.

6

Later-year data needs a caveat

The apparent 2019+ drop should be verified against source completeness before use in planning.

Built for Interactive, Public Storytelling

INTERACTIVE FILTERS

● Primary Type

Isolate any of the 33 crime categories, from Theft to Homicide, across every dashboard.

● Location Description

Narrow the view to a setting — street, residence, airport, abandoned building, and more.

● Year

Trend views additionally filter by year (2017–2022) to compare specific periods.

DESIGNED FOR THE PUBLIC

- A live feed and headline total open the story, so visitors see the current picture in seconds.
- Geo-mapping grounds abstract counts in real neighborhoods and blocks.
- Time and trend views separate 'when' from 'how much,' avoiding a single overloaded chart.
- Consistent filters across all four dashboards let one selection carry through the whole story.

CLOSING

Recommendations & Next Steps

- 1 Weight afternoon and night patrol resourcing toward the categories and blocks the dashboard flags as highest-volume.
- 2 Review why 80% of incidents close without arrest, starting with the highest-count categories: Theft and Battery.
- 3 Confirm the 2019–2022 data extract's completeness before using the year-over-year trend for planning.
- 4 Publish the dashboard to the communication website with the shared Primary Type and Location filters live for public use.

Thank you.

Crime Analysis — Course-End Project · Built in Tableau